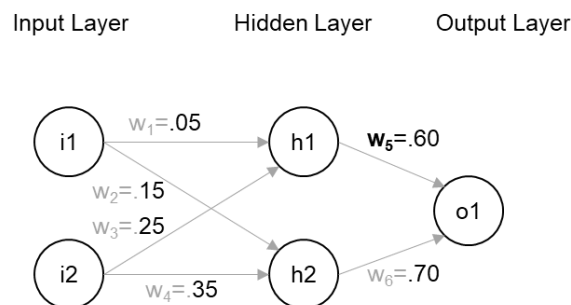


Exercise 1: Neural Networks

Consider the following neural network with two input nodes, two nodes in a single hidden layer, and one output node. The network's current weights are indicated on the edges.



- a) For input values of $i1 = 1$, $i2 = 2$, compute the nodes' output values $\Phi(z)$ as well as the intermediate z values. Please use the sigmoid activation function (see below). Assume all bias units (b) to be zero.

	z (=net)	$\Phi(z)$ (=out)
h1		
h2		
o1		

- b) Assume that the **target value was 0**. What is the total error?

- c) Now, conduct one update step for w_5 (using backward propagation). To do so, make use of the piecewise multiplicativity between total and partial derivatives and first determine the individual derivatives and values.

$$\frac{\partial E_{total}}{\partial w_5} = \frac{\partial E_{total}}{\partial out_{o1}} * \frac{\partial out_{o1}}{\partial net_{o1}} * \frac{\partial net_{o1}}{\partial w_5}$$

Partial Derivative	Formula	Value
$\frac{\partial E_{total}}{\partial out_{o1}}$		
$\frac{\partial out_{o1}}{\partial net_{o1}}$		
$\frac{\partial net_{o1}}{\partial w_5}$		

- d) Now, using gradient descent, conduct the update step. Assume a learning rate of 0.5. What is the new value of w_5 ?

Hints:

$z = \sum_i w_i x_i + b$	Derivate of Sigmoid function: $f(x) = \frac{1}{1 + e^{-x}} = \frac{e^x}{1 + e^x}$ $\frac{d}{dx} f(x) = \frac{e^x \cdot (1 + e^x) - e^x \cdot e^x}{(1 + e^x)^2}$ $\frac{d}{dx} f(x) = \frac{e^x}{(1 + e^x)^2} = f(x)(1 - f(x))$
$\phi(z) = \frac{1}{1 + e^{-z}}$	
$E_{total} = \sum \frac{1}{2} (target - output)^2$	

Exercise 2: Image Classification

Consider the datasets furniture images `scandi.zip`, `modern.zip`, and `pop.zip` (as provided through ISIS). Make yourself familiar with the online service <https://app.ximilar.com> and use it to train a model that distinguishes between the furniture styles. Which accuracy can you reach?

Example images

Scandi



Modern



Pop

